Exploratory Data Analysis

(EDA) project

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Executive Summary

87,396
Records after cleaning

40.88%
Cancellation: 365+ day lead

Transient
Highest average ADR

August
Highest booking month

- Longer lead times are associated with substantially higher cancellation risk.
- Transient customers have the highest average ADR among customer types.
- International guests have higher average ADR and longer stays than domestic guests.
- Repeat guests book much closer to arrival than new guests.
- Online TA is the largest market segment and carries the highest cancellation rate among major segments.

Dataset & Data Cleaning

119,390
Original rows

31,994
Duplicates removed

87,396
Rows after cleaning

0
Missing values after cleaning

- Removed duplicate records.
- Dropped the company column because of extensive missingness.
- Filled missing agent values with 0, country with 'Unknown', and children with 0.
- Converted reservation_status_date to datetime.
- Created total_stay and guest_origin for deeper analysis.

Booking Volume Over Time



Key finding

August has the highest number of bookings (11,257). Seasonal variation is visible across arrival months.

Useful for demand planning and staffing decisions.

Lead Time vs Booking Cancellation

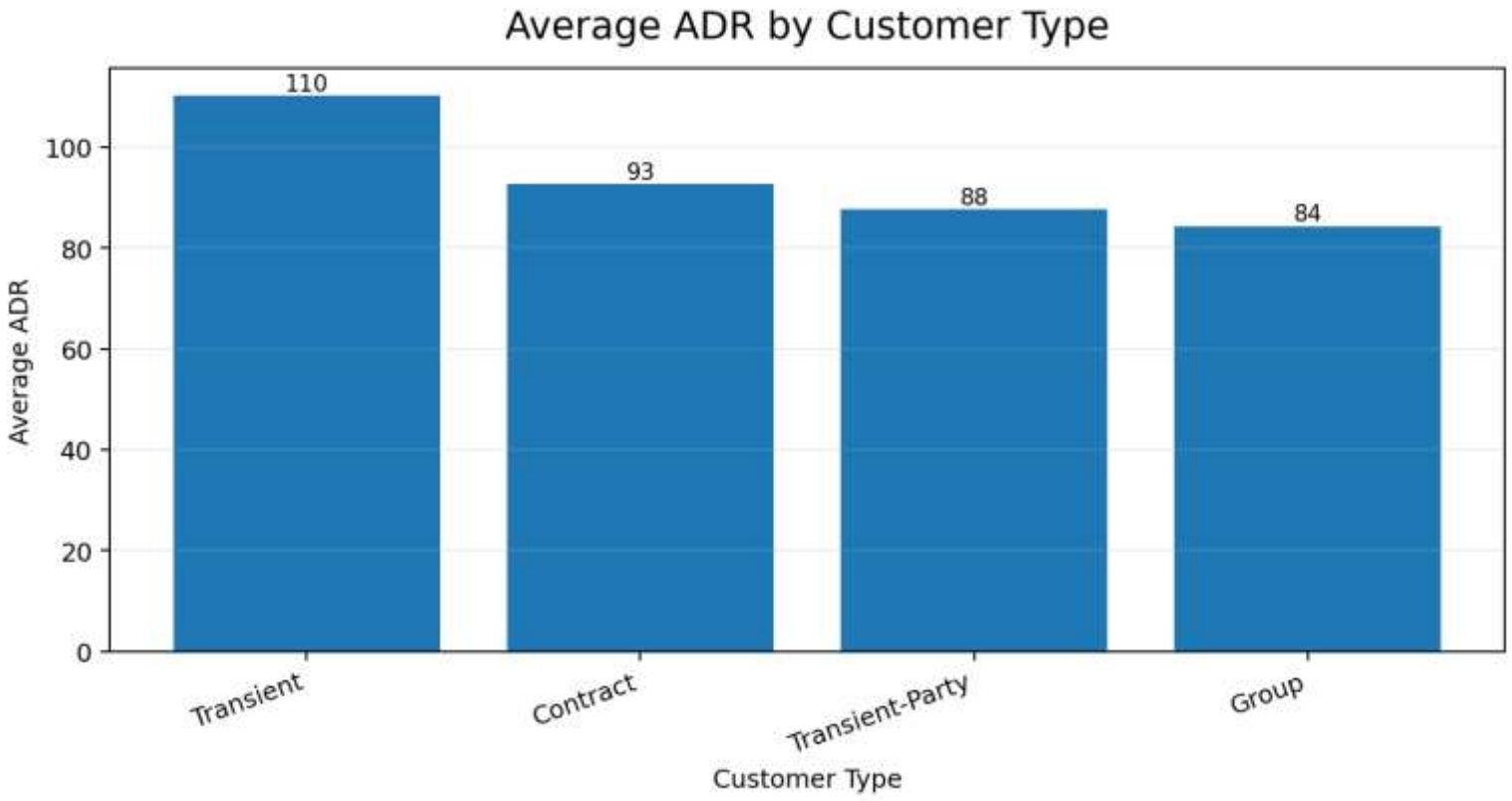


Key finding

Cancellation rises from 8.43% for 0–7 day lead times to 40.88% for 365+ day lead times.

Long-lead reservations are the clearest cancellation-risk area.

Customer Segmentation & ADR

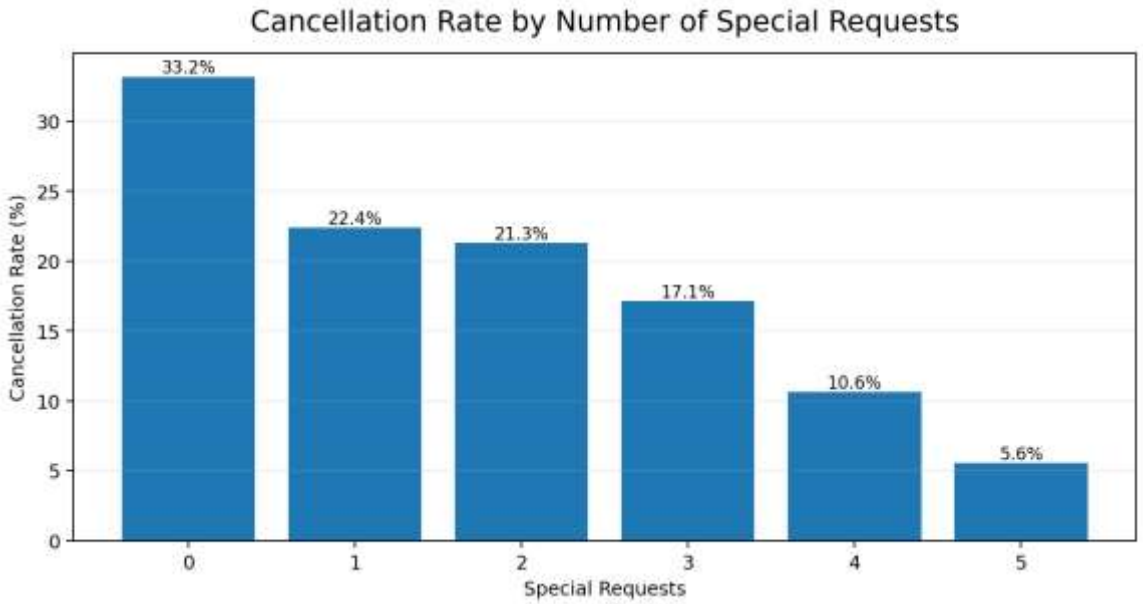


Key finding

Transient customers have the highest average ADR (110.14), followed by Contract (92.75).

Transient customers also have relatively high cancellation exposure.

Booking Behavior Signals

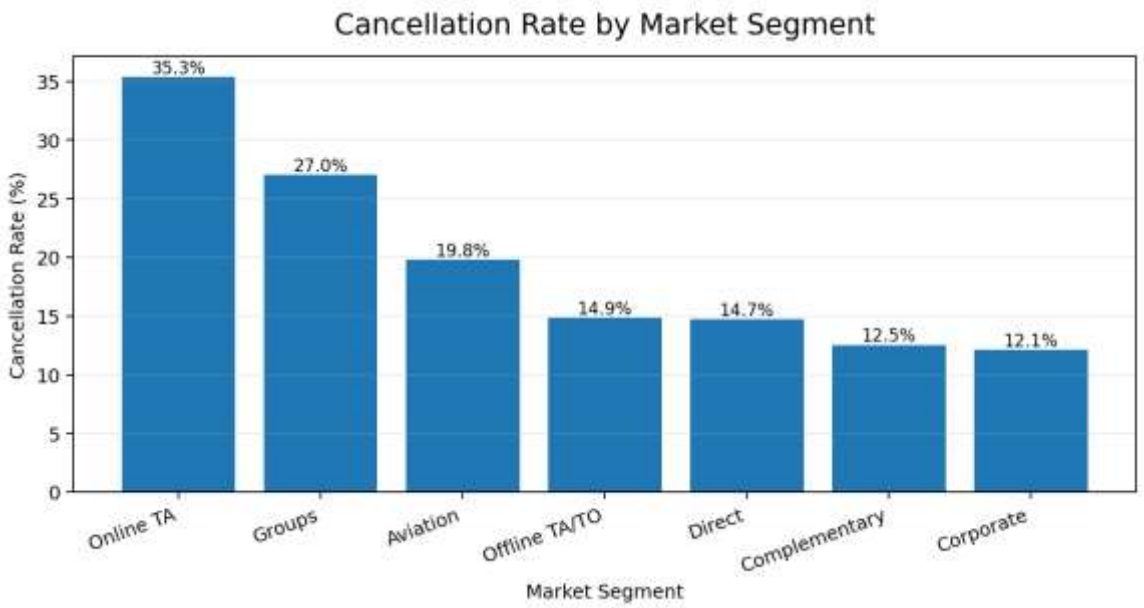


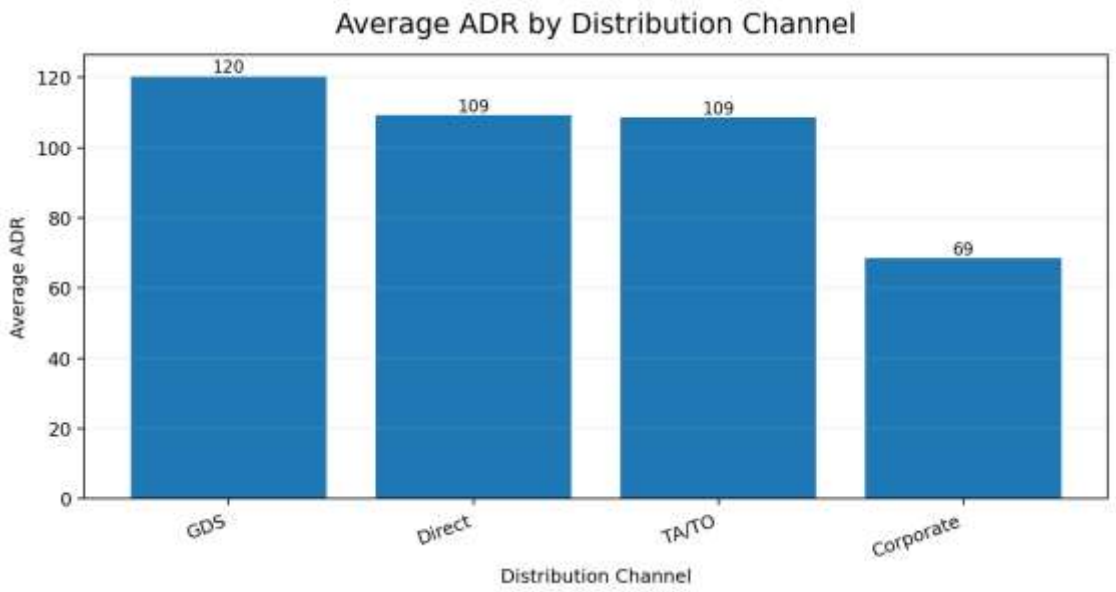
43,502
Bookings with ≥1 special request

0.70
Avg. special requests

- No special requests: 33.2% cancellation rate.
- Five special requests: 5.6% cancellation rate.
- New guests: 82.4 days average lead time.
- Repeat guests: 17.2 days average lead time.

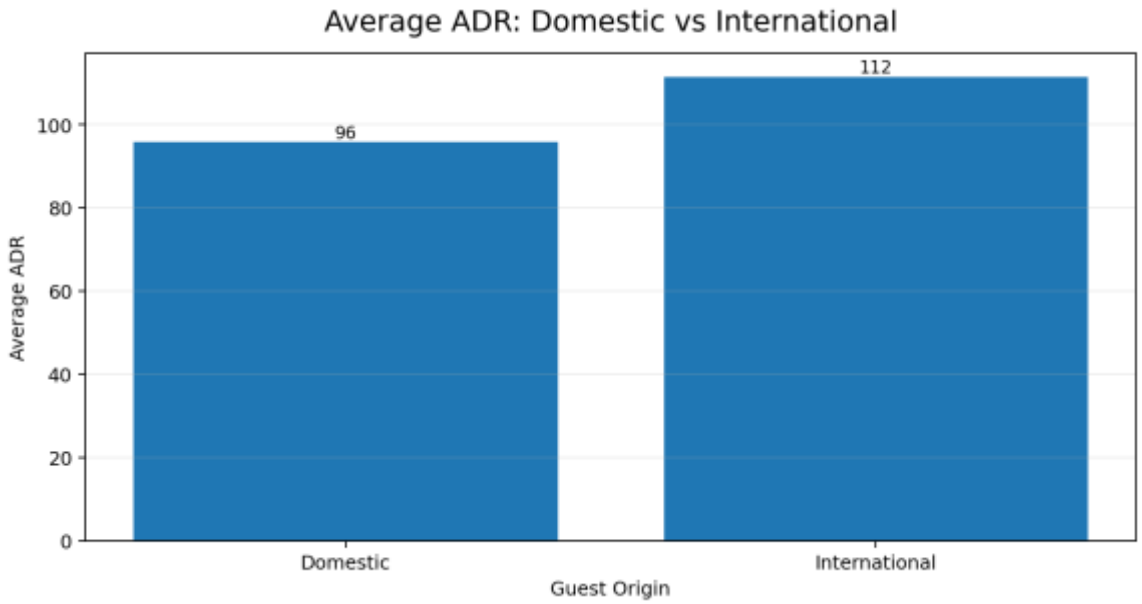
Market Segments & Distribution Channels

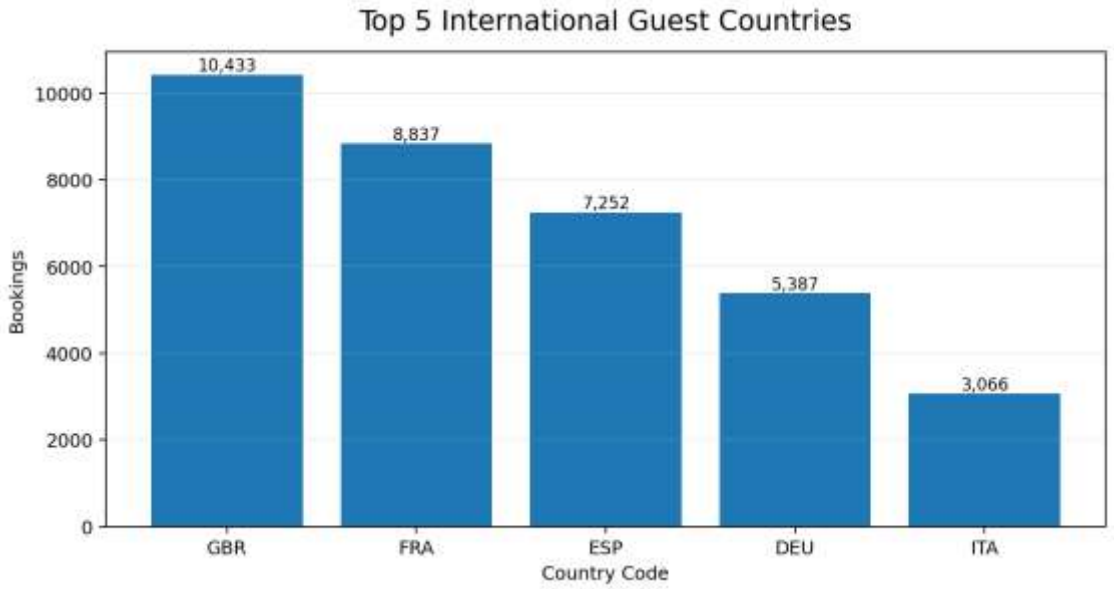




- Online TA has the highest cancellation rate among major market segments.
- GDS has the highest average ADR (120.32).
- Channel mix therefore affects both revenue quality and cancellation exposure.

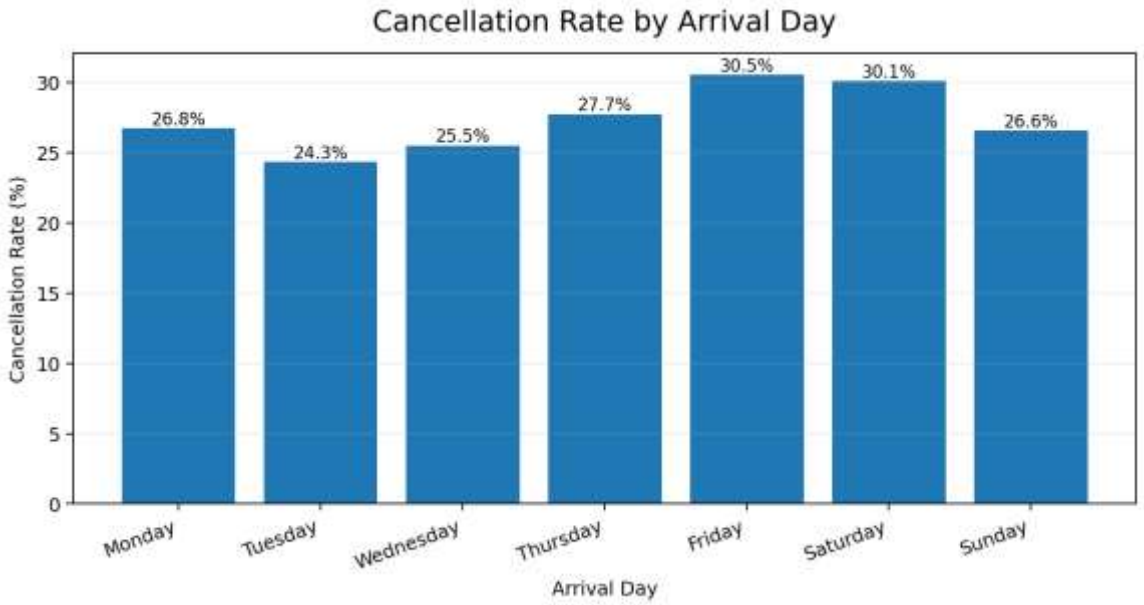
Domestic vs International Guests

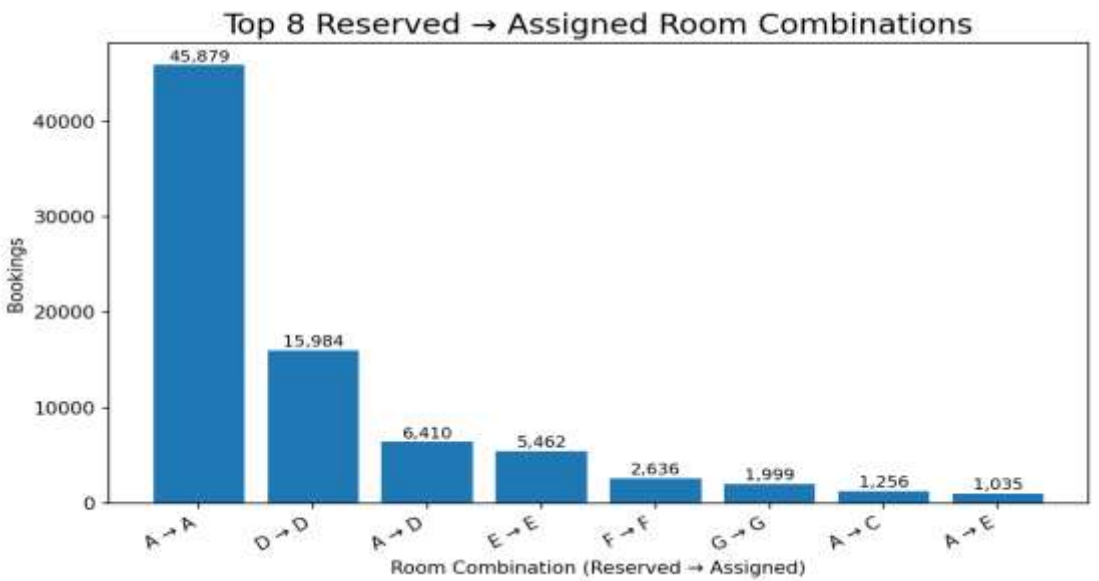




- International: ADR 111.52; stay 3.89 nights.
- Domestic: ADR 95.84; stay 3.09 nights.
- Top international markets: GBR, FRA, ESP, DEU and ITA.

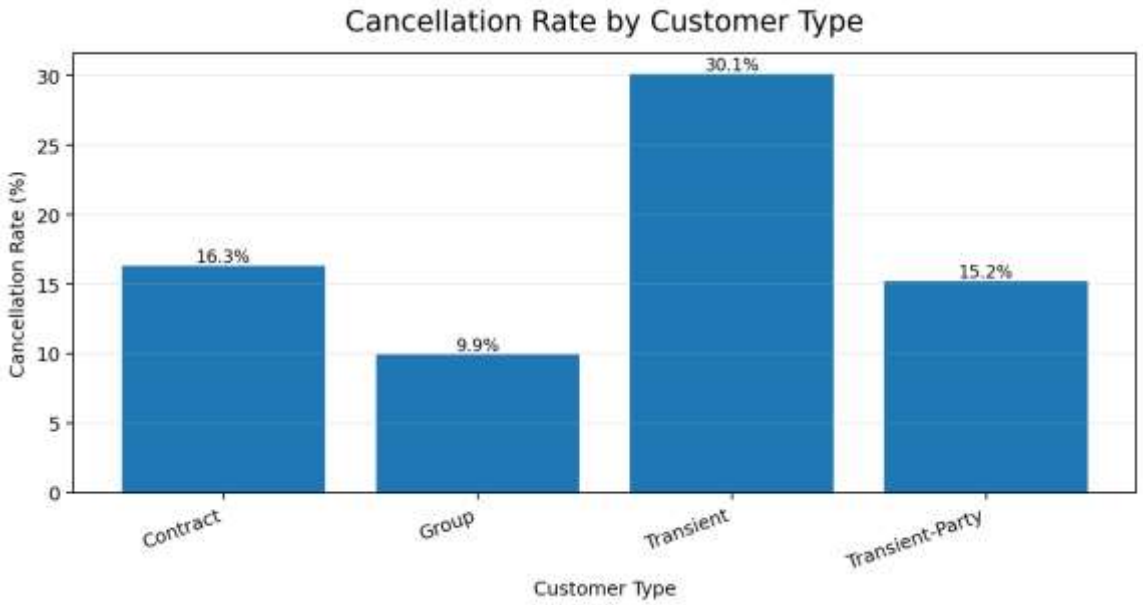
Arrival Patterns & Room Allocation





- Friday has the highest cancellation rate (30.54%), while Tuesday is lowest (24.32%).
- A → A is the most common reserved-to-assigned room combination (45,879 bookings).
- Day-of-week differences exist, but they are not extreme.

Customer Segment Comparison



- Transient: ADR 110.14; cancellation 30.1%.
- Contract: ADR 92.75; average stay 6.0 nights.
- Group: lowest cancellation rate (9.9%).
- Transient-Party: ADR 87.68; moderate cancellation risk.

Actionable Insights

1. Target long-lead bookings with stronger confirmation reminders and appropriate cancellation policies.

2. Monitor Online TA bookings closely because of higher cancellation exposure.

3. Protect and grow the transient segment because it generates the highest average ADR.

4. Strengthen repeat-guest programs; repeat guests book much closer to arrival.

5. Tailor campaigns toward high-volume international markets such as GBR, FRA and ESP.

6. Use arrival-day patterns for staffing and inventory planning, while recognizing the differences are moderate.

7. Use special-request behavior as a customer-engagement signal because higher request counts are associated with lower cancellation.

8. Compare channel-level ADR to improve the mix of higher-value bookings.

Objective Findings: Questions 1–5

105.72 vs 70.10
Avg lead time (days): cancelled vs not

11,257
Bookings in August (highest month)

110.14
Highest avg ADR — Transient

43,502
Bookings with ≥1 special request

Q1: Cancelled bookings average 105.72 days of lead time versus 70.10 days for non-cancelled bookings.

Q2: August records the highest booking volume with 11,257 bookings.

Q3: ADR distributions differ across customer types; Transient has the highest average ADR (110.14).

Q4: 43,502 bookings contain at least one special request; the average is 0.70 requests per booking.

Q5: Cancellation rates are highest for Online TA (35.3%) among the major market segments.

Objective Findings: Questions 6–10

120.32
Highest-ADR distribution channel — GDS

0.318
Lead time ↔ total stay correlation

9,103
Bookings with children and/or babies

45,879
Most common room match: A → A

Q6: GDS has the highest average ADR (120.32) among distribution channels.

Q7: Lead time and length of stay have a positive correlation of 0.318.

Q8: Children and/or babies are present in 9,103 bookings.

Q9: Repeat guests book much closer to arrival (17.2 days) than new guests (82.4 days).

Q10: A → A is the most common reserved/assigned room combination (45,879 bookings).

Final Conclusion

The EDA shows that hotel booking behavior is strongly influenced by lead time, customer type, market segment and guest origin. The clearest pattern is the rise in cancellation risk as bookings are made farther in advance. Transient customers generate the highest ADR, international guests generally show higher ADR and longer stays than domestic guests, and repeat guests book much closer to arrival. These findings can support better cancellation controls, revenue management, customer targeting and operational planning.

Thank you